

MATH 2R03 - Assignment 2

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Question 1

Let V be a vector space over a field $\mathbb{F}$, and let $v_1, \dots, v_m$ be a list of vectors in V . For $k = 1, \dots, m$ set $w_k := v_1 + \dots + v_k$. Show that $\text{span}(w_1, \dots, w_m) = \text{span}(v_1, \dots, v_m)$.

Solution:

To prove that $\text{span}(w_1, \dots, w_m) = \text{span}(v_1, \dots, v_m)$, we must show that $\text{span}(w_1, \dots, w_m) \subseteq \text{span}(v_1, \dots, v_m)$ and $\text{span}(v_1, \dots, v_m) \subseteq \text{span}(w_1, \dots, w_m)$.

(I) Prove $\text{span}(w_1, \dots, w_m) \subseteq \text{span}(v_1, \dots, v_m)$.

Let $u \in \text{span}(w_1, \dots, w_m)$. By definition of span, we know that there exists $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ such that:

$$\lambda_1 w_1 + \dots + \lambda_m w_m = u$$

However, we know that $w_k = v_1 + \dots + v_k$ by its definition. Therefore, $\lambda_k w_k = \lambda_k(v_1 + \dots + v_k) = \lambda_k v_1 + \dots + \lambda_k v_k$. Therefore, we have that $\lambda_k w_k + \lambda_{k+1} w_{k+1} + \dots + \lambda_m w_m = (\lambda_k + \lambda_{k+1} + \dots + \lambda_m)v_k + (\lambda_{k+1} + \dots + \lambda_m)v_{k+1} + \dots + \lambda_m v_m$ and hence:

$$\begin{aligned} \lambda_1 v_1 + \dots + \lambda_m(v_1 + \dots + v_m) &= u \implies \\ \lambda_1 v_1 + \dots + \lambda_m v_1 + \dots + \lambda_m v_m &= u \implies \\ (\lambda_1 + \dots + \lambda_m)v_1 + \dots + \lambda_m v_m &= u \end{aligned}$$

Thus, since $(\lambda_k + \dots + \lambda_m) \in \mathbb{F}$, letting $\alpha_k \in \mathbb{F}$ such that $\alpha_k = \sum_{i=k}^m \lambda_i$, we have that:

$$\alpha_1 v_1 + \dots + \alpha_m v_m = u$$

And thus $u \in \text{span}(v_1, \dots, v_m)$. Therefore, $\text{span}(w_1, \dots, w_m) \subseteq \text{span}(v_1, \dots, v_m)$.

(II) Prove $\text{span}(v_1, \dots, v_m) \subseteq \text{span}(w_1, \dots, w_m)$.

Let $u \in \text{span}(v_1, \dots, v_m)$. By definition of span, we know that there exists $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ such that:

$$\lambda_1 v_1 + \dots + \lambda_m v_m = u$$

Now, let $\alpha_k \in \mathbb{F}$ such that $\alpha_m = \lambda_m, \alpha_{m-1} = \lambda_{m-1} - \lambda_m, \dots, \alpha_1 = \lambda_1 - \lambda_2 - \dots - \lambda_m$. By definition of w_k , we have that:

$$\begin{aligned}\alpha_1 w_1 + \dots + \alpha_m w_m &= \alpha_1 v_1 + \dots + \alpha_m (v_1 + \dots + v_m) \\ &= \alpha_1 v_1 + \dots + \alpha_m v_1 + \dots + \alpha_m v_m\end{aligned}$$

And by our construction of α_k , we have that:

$$\begin{aligned}\alpha_1 v_1 + \dots + \alpha_m v_1 + \dots + \alpha_m v_m &= (\lambda_1 - \dots - \lambda_m) v_1 + \dots + \lambda_m v_1 + \dots + \lambda_m v_m \\ &= \lambda_1 v_1 - \dots - \lambda_m v_1 + \dots + \lambda_m v_1 + \dots + \lambda_m v_m \\ &= \lambda_1 v_1 + \dots + \lambda_m v_m \\ &= u\end{aligned}$$

And thus $u \in \text{span}(w_1, \dots, w_m)$. Therefore, $\text{span}(v_1, \dots, v_m) \subseteq \text{span}(w_1, \dots, w_m)$.

Therefore, since $\text{span}(w_1, \dots, w_m) \subseteq \text{span}(v_1, \dots, v_m)$ and $\text{span}(v_1, \dots, v_m) \subseteq \text{span}(w_1, \dots, w_m)$, $\text{span}(v_1, \dots, v_m) = \text{span}(w_1, \dots, w_m)$.

Question 2

Prove or give a counterexample: $v_1, \dots, v_m$ is a linearly independent list of vectors in V , then $5v_1 - 4v_2, v_2, \dots, v_m$ is linearly independent.

Solution:

Let $\lambda_1, \dots, \lambda_m \in \mathbb{F}$. We can see that:

$$\begin{aligned}\lambda_1(5v_1 - 4v_2) + \lambda_2v_2 + \dots + \lambda_mv_m &= 0 \implies \\ 5\lambda_1v_1 - 4\lambda_1v_2 + \lambda_2v_2 + \dots + \lambda_mv_m &= 0 \implies \\ 5\lambda_1v_1 + (\lambda_2 - 4\lambda_1)v_2 + \dots + \lambda_mv_m &= 0\end{aligned}$$

Now since $v_1, \dots, v_m$ are linearly independent, we know that this implies $5\lambda_1 = (\lambda_2 - 4\lambda_1) = \dots = \lambda_m = 0$. Thus, if $5\lambda_1 = 0$, then $\lambda_1 = 0$, and if $\lambda_1 = 0$, then $\lambda_2 - 4\lambda_1 = 0$ implies $\lambda_2 = 0$. Thus, we have that:

$$\lambda_1(5v_1 - 4v_2) + \lambda_2v_2 + \dots + \lambda_mv_m = 0$$

Implies $\lambda_1 = \dots = \lambda_m = 0$ and hence $5v_1 - 4v_2, v_2, \dots, v_m$ is linearly independent.

Question 3

Let $v_1, \dots, v_m$ be a linearly independent list of vectors in V , and let $w \in V$. Show that if the list $v_1, \dots, v_m, w$ is a linearly independent list if and only if $w \notin \text{span}(v_1, \dots, v_m)$.

Solution:

We will prove this by showing the forward and reverse directions separately.

(I) Forward direction.

Assume, for the purpose of contradiction, that $v_1, \dots, v_m, w$ is a linearly independent list but that $w \in \text{span}(v_1, \dots, v_m)$. Then by definition of span, we know there exists $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ such that:

$$\lambda_1 v_1 + \dots + \lambda_m v_m = w$$

However, note that this would imply that:

$$\lambda_1 v_1 + \dots + \lambda_m v_m + (-w) = 0$$

Which implies that $v_1, \dots, v_m, w$ cannot be a linearly independent list of vectors since w has a non-zero coefficient. Therefore, by contradiction, $v_1, \dots, v_m, w$ being a linearly independent list implies that $w \notin \text{span}(v_1, \dots, v_m)$.

(II) Reverse direction

Once again, assume for the purposes of contradiction that $w \notin \text{span}(v_1, \dots, v_m)$ but that both $v_1, \dots, v_m$ and $v_1, \dots, v_m, w$ are linearly dependent lists of vectors. Then, we know there exists $\lambda_1, \dots, \lambda_m, \lambda_{m+1} \in \mathbb{F}$ where some $\lambda_k \neq 0$ such that:

$$\lambda_1 v_1 + \dots + \lambda_m v_m + \lambda_{m+1} w = 0$$

However, this would imply that:

$$\lambda_1 v_1 + \dots + \lambda_m v_m = -\lambda_{m+1} w$$

Now, if $\lambda_{m+1} \neq 0$, then we have that:

$$-\frac{\lambda_1}{\lambda_{m+1}}v_1 + \dots - \frac{\lambda_m}{\lambda_{m+1}}v_m = w$$

And thus, $w \in \text{span}(v_1, \dots, v_m)$ which would be a contradiction. If $\lambda_{m+1} = 0$, then we would have that:

$$\begin{aligned}\lambda_1 v_1 + \dots + \lambda_m v_m &= -\lambda_{m+1} w \implies \\ \lambda_1 v_1 + \dots + \lambda_m v_m &= 0\end{aligned}$$

And since at least one $\lambda_k \neq 0$, that would imply that $v_1, \dots, v_m$ is not a linearly independent list of vectors, which would also be a contradiction. Therefore, $v_1, \dots, v_m$ being an independent list of vectors and $w \notin \text{span}(v_1, \dots, v_m)$ implies that $v_1, \dots, v_m, w$ is a linearly independent list of vectors.

Thus, for a linearly independent list of vectors $v_1, \dots, v_m$, we have that $v_1, \dots, v_m, w$ is also a linearly independent list of vectors if and only if $w \notin \text{span}(v_1, \dots, v_m)$.

Question 4

Let $\mathbb{F}$ be a field. For $n \in \mathbb{N}$, let:

$$\mathbb{F}[x]_n := \{ p \in \mathbb{F}[x] : \deg(p) \leq n \}$$

(a) Prove that $\mathbb{F}[x]_n$ is a subspace of $\mathbb{F}[x]$.

Solution:

(b) Explain why there does not exist a list of six polynomials that is linearly independent in $\mathbb{F}[x]_4$

Solution:

We know that any list of six polynomials cannot be a linearly independent list in $\mathbb{F}[x]_4$ since a list of five polynomials forms a basis of $\mathbb{F}[x]_4$, and hence any list of six vectors in $\mathbb{F}[x]_4$ cannot be linearly independent.

(c) Explain why no list of four polynomials spans $\mathbb{F}[x]_4$.

Solution:

We know that any list of four polynomials cannot span $\mathbb{F}[x]_4$ since a list of five polynomials forms a basis of $\mathbb{F}[x]_4$, and hence any list of four vectors cannot span $\mathbb{F}[x]_4$.

Question 5

Consider the set $F_3 := \{0, 1, 2\}$ together with the addition $a \oplus b := (a + b) \bmod 3$ and the multiplication $a \otimes b := (ab) \bmod 3$.

(a) Verify that $\oplus$ and $\otimes$ satisfy the field axioms; only submit here the proof of distributivity.

Solution:

(b) List all functions in the vector space $F_3^{F_3}$ (how many are there?), and determine which of these are polynomial functions.

Solution:

There are a total of 27 functions in the vector space $F_3^{F_3}$:

$$f(2) = 0 \quad f(1) = 0 \quad f(0) = 0 \quad (1)$$

$$f(2) = 0 \quad f(1) = 0 \quad f(0) = 1 \quad (2)$$

$$f(2) = 0 \quad f(1) = 0 \quad f(0) = 2 \quad (3)$$

$$f(2) = 0 \quad f(1) = 1 \quad f(0) = 0 \quad (4)$$

$$f(2) = 0 \quad f(1) = 1 \quad f(0) = 1 \quad (5)$$

$$f(2) = 0 \quad f(1) = 1 \quad f(0) = 2 \quad (6)$$

$$f(2) = 0 \quad f(1) = 2 \quad f(0) = 0 \quad (7)$$

$$f(2) = 0 \quad f(1) = 2 \quad f(0) = 1 \quad (8)$$

$$f(2) = 0 \quad f(1) = 2 \quad f(0) = 2 \quad (9)$$

$$f(2) = 1 \quad f(1) = 0 \quad f(0) = 0 \quad (10)$$

$$f(2) = 1 \quad f(1) = 0 \quad f(0) = 1 \quad (11)$$

$$f(2) = 1 \quad f(1) = 0 \quad f(0) = 2 \quad (12)$$

$$f(2) = 1 \quad f(1) = 1 \quad f(0) = 0 \quad (13)$$

$$f(2) = 1 \quad f(1) = 1 \quad f(0) = 1 \quad (14)$$

$$f(2) = 1 \quad f(1) = 1 \quad f(0) = 2 \quad (15)$$

$$f(2) = 1 \quad f(1) = 2 \quad f(0) = 0 \quad (16)$$

$$f(2) = 1 \quad f(1) = 2 \quad f(0) = 1 \quad (17)$$

$$f(2) = 1 \quad f(1) = 2 \quad f(0) = 2 \quad (18)$$

$$f(2) = 2 \quad f(1) = 0 \quad f(0) = 0 \quad (19)$$

$$f(2) = 2 \quad f(1) = 0 \quad f(0) = 1 \quad (20)$$

$$f(2) = 2 \quad f(1) = 0 \quad f(0) = 2 \quad (21)$$

$$f(2) = 2 \quad f(1) = 1 \quad f(0) = 0 \quad (22)$$

$$f(2) = 2 \quad f(1) = 1 \quad f(0) = 1 \quad (23)$$

$$f(2) = 2 \quad f(1) = 1 \quad f(0) = 2 \quad (24)$$

$$f(2) = 2 \quad f(1) = 2 \quad f(0) = 0 \quad (25)$$

$$f(2) = 2 \quad f(1) = 2 \quad f(0) = 1 \quad (26)$$

$$f(2) = 2 \quad f(1) = 2 \quad f(0) = 2 \quad (27)$$

All of them, in fact, have a polynomial representation. Letting f be any of the functions listed above where $f(0) = a$, $f(1) = b$, and $f(2) = c$, we can express $f(x)$ as:

$$f(x) = \frac{a}{2}(x-1)(x-2) - bx(x-2) + \frac{c}{2}x(x-1)$$

Since:

$$\begin{aligned} f(0) &= \frac{a}{2}(0-1)(0-2) - b(0)(0-2) + \frac{c}{2}(0)(0-1) \\ &= \frac{a}{2}(-1)(-2) \\ &= \frac{a}{2}(2) \\ &= a \end{aligned}$$

$$\begin{aligned} f(1) &= \frac{a}{2}(1-1)(1-2) - b(1)(1-2) + \frac{c}{2}(1)(1-1) \\ &= -b(1)(-1) \\ &= -b(-1) \\ &= b \end{aligned}$$

$$\begin{aligned} f(2) &= \frac{a}{2}(2-1)(2-2) - b(2)(2-2) + \frac{c}{2}(2)(2-1) \\ &= \frac{c}{2}(2)(1) \\ &= \frac{c}{2}(2) \\ &= c \end{aligned}$$

Thus, since such an f is clearly polynomial and can be used to construct all the functions listed above, all the functions in the vector space $F_3^{F_3}$ are polynomial.

(c) Give a linearly independent list of polynomial functions that spans $\mathcal{P}(F_3)$.

Solution: